

Introduction to Artificial Intelligence

Week 1 Assignment • Task 1 Report

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that focuses on building machines and software that can perform tasks which normally need human intelligence. This includes things like understanding language, recognizing images, making decisions, and learning from past experience. In simple words, AI tries to make computers 'think' and 'act' in a smart way, similar to how a human would.

2. Types of AI (ANI, AGI, ASI)

AI is generally divided into three types based on how capable it is:

- **ANI (Artificial Narrow Intelligence):** AI that is good at only one specific task, like a chatbot, a spam filter, or a voice assistant. This is the only type of AI that exists today.
- **AGI (Artificial General Intelligence):** AI that can understand, learn and perform any intellectual task a human can. This does not exist yet, it is still a research goal.
- **ASI (Artificial Super Intelligence):** AI that would be smarter than humans in every field. This is only a hypothetical/future concept discussed in research and science fiction.

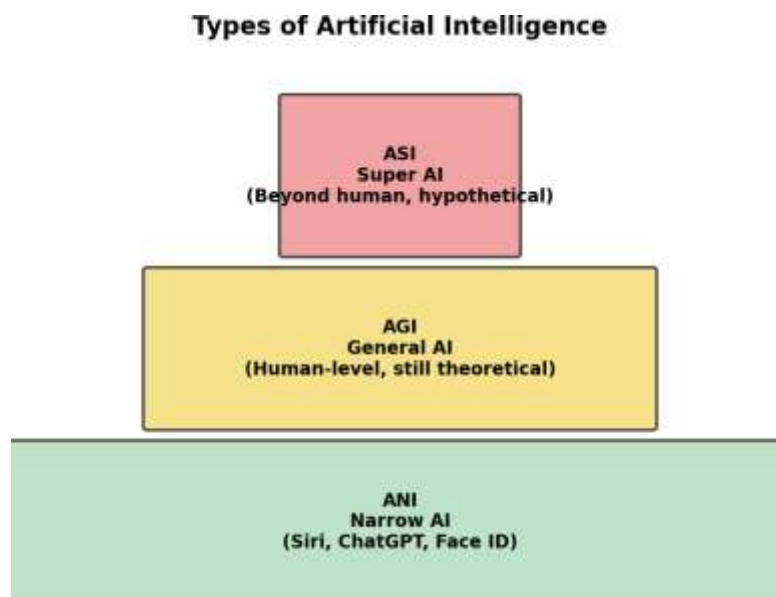


Fig 1: The three types of AI, from narrow (existing today) to super (hypothetical).

3. AI vs Machine Learning vs Deep Learning

These three terms are related but not the same thing. AI is the overall goal of making machines intelligent. Machine Learning (ML) is a subset of AI where the machine learns patterns from data instead of being explicitly programmed with rules. Deep Learning (DL) is a further subset of ML that uses multi-layer neural networks, and is especially good at tasks like image recognition and natural language processing.

AI vs Machine Learning vs Deep Learning

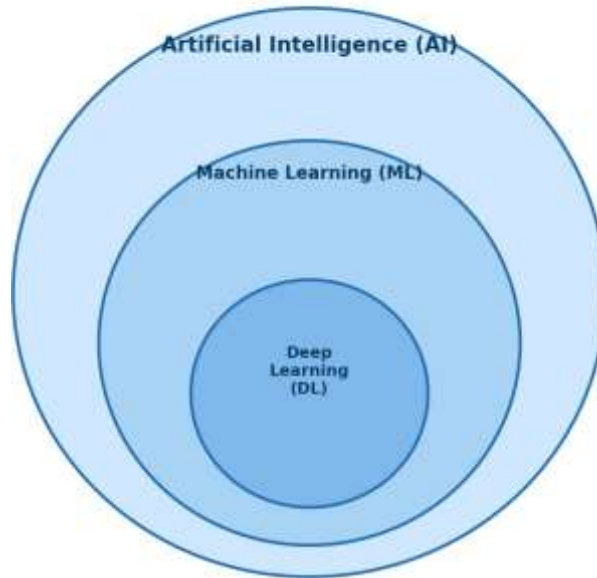


Fig 2: Deep Learning is inside Machine Learning, which is inside AI.

Aspect	AI	Machine Learning	Deep Learning
Definition	Making machines act smart	Learning from data	Learning using neural networks
Scope	Broadest	Subset of AI	Subset of ML
Example	Any smart system	Spam detection	Image / speech recognition

4. Real-Life Applications of AI

AI is already part of everyday life. A few common examples are: voice assistants such as Siri, Alexa and Google Assistant; recommendation systems used by Netflix, YouTube and Amazon; face recognition used for phone unlocking and security; self-driving car technology; AI chatbots for customer support; and AI-based tools in healthcare that help detect diseases from scans.

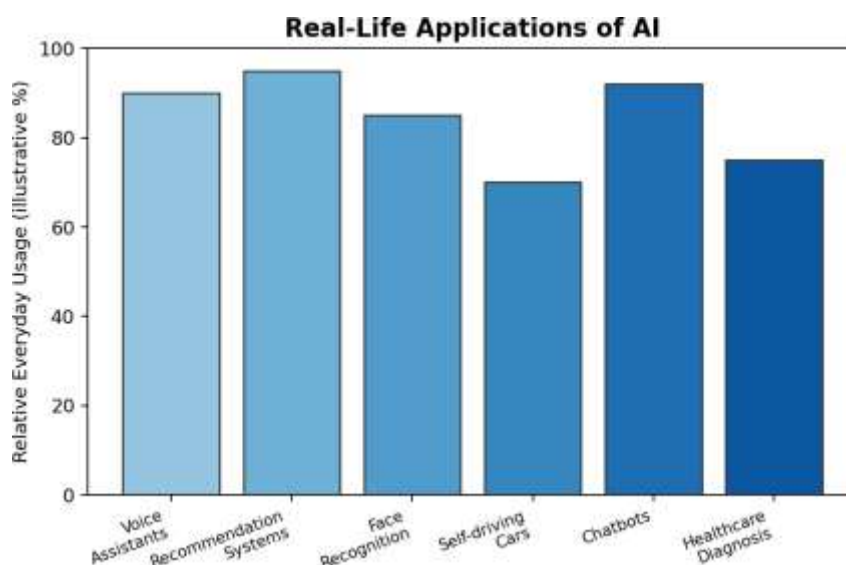


Fig 3: Some everyday applications of AI.

5. Advantages and Limitations of AI

Advantages	Limitations
Can work 24/7 without getting tired	Very expensive to build and train
Reduces human error in repetitive tasks	Needs large amounts of good quality data
Can process huge amounts of data quickly	Lacks common sense and real understanding
Helps in fields like healthcare, finance, education	Can be biased if training data is biased
Automates boring / repetitive work	Raises job-replacement and ethical concerns